

3 (a) resonance B1 [1]

(b) $Pt = mc \Delta\theta$ C1

$$750 \times 2 \times 60 = 0.28 \times c \times (98 - 25)$$
C1

$$c = 4400 \text{ J kg}^{-1} \text{ K}^{-1}$$
A1 [3]

(use of $\Delta\theta = 73 + 273$ max. 1/3)

(use of $t = 2\text{ s}$ not 120 s max. 2/3)

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- (c) e.g. some microwave leakage from the cooker
e.g. container for the water is also heated
(any sensible suggestion)

B1 [1]